Yulong Liu

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Research Interests

AI methods & discovery: Differentiable programming, hybrid neural PDE solvers, uncertainty-aware scientific AI, and AI for Science and scientific discovery.

Engineering: Homogenization and multiscale modeling, coupled multiphysics simulation, and subsurface energy systems, including geothermal reservoirs and hydraulic fracturing.

Education

Ph.D. in Earth Science – Cornell University 2024 – Present
Minors: Computer Science and Scientific Computing
B.S. in Mining Engineering – Northeastern University (CN) 2024

Research Experience

Ph.D. Candidate – Dept. of Earth and Atmospheric Sciences, Cornell University Aug. 2024 – Present
Advisor: Chloé Arson *Ithaca, NY*
Project: Scientific AI and scientific discovery for multiscale, multiphysics geomechanics and subsurface energy systems.

- Develop coupled thermo-hydro-mechanical models in MOOSE for subsurface energy systems, including geothermal reservoirs and hydraulic fracturing.
- Study differentiable programming and hybrid neural PDE solvers, combining physics-informed networks, implicit neural representations, and neural operators with numerical simulation.
- Investigate uncertainty-aware surrogate modeling for geothermal systems, with an emphasis on uncertainty propagation and the reliability of multiphysics predictions.
- Lead a cross-institution project on PDE-grounded verification of the physical correctness of LLM-generated multiphysics simulation code.
- Study homogenization and multiscale transport in heterogeneous porous media; use AI for scientific discovery to learn interpretable effective permeability laws.

Undergraduate Researcher – Northeastern University (CN) Sep. 2020 – Jun. 2024
B.S. in Mining Engineering *Shenyang, China*
Thesis: A Unified Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning

- Developed a unified deep-learning model for microseismic signal classification and arrival-time picking, integrating tasks typically handled in separate workflows.
- Conducted experimental and computational research on hydraulic fracturing, triaxial rock testing, and tunnel boring machines; received the Outstanding Thesis Award and National Scholarship.

Grants, Proposals, and Computational Awards

Homogenization and Machine Learning for Multiscale Porous Media Flow 2026
National Science Foundation (NSF), ACCESS program (EES260208). PIs: ***Yulong Liu*** *and C. Arson.*

Learned Uncertainty-Propagation and World Models for Enhanced Geothermal Reservoirs 2026
National Science Foundation (NSF), ACCESS program. PIs: ***Yulong Liu*** *and C. Arson. Total allocation and Arson group's share: 1.5M CPU hours.*

AI-aided Computational Geomechanics 2026
Empire AI Alpha+Beta. PIs: Y.C. Han, ***Yulong Liu****, A. Tristani, and C. Arson. Total allocation and Arson group's share: 5.5K GPU hours.*

NASA FINESST Proposal – Title Withheld During Anonymous Review 2026
NASA ROSES-25 F.5 FINESST. Under review. Future Investigator: ***Yulong Liu****; PI and faculty mentor: C. Arson.*

Publications

Selected Papers

- **Yulong Liu**, Jonah Botvinick-Greenhouse, Yunan Yang, Chloé Arson. *Operator Learning Surrogate Modeling of Hydraulically Fractured Geothermal Injection–Production Systems: A Cornell Case Study.* ARMA US Rock Mechanics/Geomechanics Symposium, Tucson, AZ, 2026.
- **Yulong Liu**, Chloé Arson. *A Physics-Informed Neural Network for Pressurized Cavities of Arbitrary Shape in Heterogeneous Rock. Rock Mechanics and Rock Engineering* (2026). doi:10.1007/s00603-026-05882-5.
- M. Belachew, **Yulong Liu**, J. D. Frost, Chloé Arson. *Numerical Assessment of Plasticity Development and Energy Expenditure of Ant-Like Microtunnelling. Tunnelling and Underground Space Technology*, 172, 107501, 2026.
- **Yulong Liu**[†], Chloé Arson. *Physics-Informed Neural Network Surrogate Modeling of Pressurized Cavity in Homogeneous and Bilayered Media.* ARMA US Rock Mechanics/Geomechanics Symposium, D022S018R006, 2025.
- Zhenghan Song, **Yulong Liu**[*,†], et al. *Your Simulation Runs but Solves the Wrong Physics: PDE-Grounded Intent Verification for LLM-Generated Multiphysics Simulation Code.* arXiv:2605.09360. Under review at NeurIPS 2026.
- Zhenghan Song, Yunyi Li, **Yulong Liu**[†]. *Prefix-Safe Bayesian Belief Tracking for LLM Reasoning Reliability: Separating Calibration from Ranking.* arXiv:2605.27712. Under review at ACL ARR 2026.

Thesis

- **Yulong Liu**. *A Unified Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning.* Undergraduate thesis, Northeastern University, 2024.

[*] Co-first author [†] Corresponding author

Conferences and Presentations

Poster presentation, 60th US Rock Mechanics/Geomechanics Symposium (ARMA) Jun. 2026
American Rock Mechanics Association, Tucson, AZ, USA

Invited talk, 20th Phase Field Methods Workshop May 2026
Center for Hierarchical Materials Design, Northwestern University, Evanston, IL, USA

Poster presentation, Cornell CEE Graduate Research Symposium Apr. 2026
Cornell University, Ithaca, NY, USA

Invited seminar: "Physics-Informed AI Optimization of Heterogeneous Rock Systems" Dec. 2025
Southwest Petroleum University seminar series, Chengdu, China

Poster presentation, 59th US Rock Mechanics/Geomechanics Symposium (ARMA) Jun. 2025
American Rock Mechanics Association, Santa Fe, NM, USA

Fellowships and Awards

Estwing Hammer Award – Most Outstanding EAS Graduate Student of 2024, Cornell University 2025
Cornell University Travel Grant, Cornell University 2025, 2026
Outstanding Thesis, Northeastern University 2024
National Scholarship, Ministry of Education, China 2024
First Class Scholarship, Northeastern University 2023
Autumn Scholarship, Northeastern University 2022
Golden Seed Scholarship, Northeastern University 2022

Academic Service

- Session Chair, GS03 Computational Geomechanics, Coupled Thermo-Hydro-Mechanical-Chemical (THMC) Processes, and Next-Generation Modeling, Geothermal Track, *61st U.S. Rock Mechanics/Geomechanics Symposium (ARMA 2027)*, Austin, TX, June 2027
- Reviewer, *Rock Mechanics and Rock Engineering*
- Reviewer, *Underground Space*
- Reviewer, *International Conference on Artificial Neural Networks*
- Reviewer, *Geothermal Rising Conference*

Skills

Programming & Scientific Computing: Python, C++, R, MOOSE finite element framework

Modeling & Simulation: finite element modeling, coupled multiphysics simulation, poromechanics, computational geomechanics, CAD (e.g., AutoCAD, SolidWorks)

Subsurface & Reservoir: geothermal reservoirs, hydraulic fracturing, injection–production modeling, homogenization and multiscale transport in porous media

Scientific Machine Learning: differentiable programming, hybrid neural PDE solvers, uncertainty-aware scientific AI, AI for Science and scientific discovery; physics-informed networks, neural operators, and implicit neural representations

LLMs for Simulation: using large language models to accelerate simulation-software workflows and to verify the physical correctness of generated simulation code

Experimental Methods: rock mechanics testing (e.g., triaxial compression), laboratory instrumentation

Currently Learning: phase-field modeling (fracture/multiphysics), multiphase flow in porous media

Scientific Writing: LaTeX / Overleaf

Professional Memberships

- Member, American Rock Mechanics Association (ARMA), 2025–Present
- Member, International Society for Rock Mechanics and Rock Engineering (ISRM), 2025–Present
- Member, Chinese Society for Rock Mechanics and Engineering (CSRME), 2024–Present
- Member, Geothermal Rising, 2025–Present